Architecting a Universal CCSDS Spacecraft TT&C and Ground EGSE Transceiver in SystemVerilog

Engineering Architecture, First-Principles Mathematical Derivations, All-Digital Baseband RF Modem DSP, Zero-Trust Space Security, and S-Band (2025–2110 / 2200–2290 MHz) RF Cascade & Link Physics

Monograph Attribute	Technical Scope & Verification Baseline
Document ID & Companion	`ENG-CCSDS-TTC-v2.40.3` (Volume II) — Companion to **Volume I: Product Guide & Hardware Integration Manual** (`PG-CCSDS-TTC-v2.40.3`)
Target Transceiver Core	`ccsds_sat_ttc_top` (`CORE_ID_VER = 32'hCC5D_0230`, IEEE 1800-2017 Synthesizable SystemVerilog)
Hardware Independence	**100% Vendor-Agnostic Core RTL (`0 BRAM / 0 DSP`)**: Uses inferred Distributed LUTRAM (`RAM64M`, 1,200 LUTs) + Flip-Flops; free of proprietary FPGA hard macros or converter-specific assumptions
Architectural Personas	`ROLE_MODE = 0` (Spacecraft Flight Core), `ROLE_MODE = 1` (Synthesizable Hardware Ground EGSE), `ROLE_MODE = 2` (Symmetric Crosslink & On-Orbit BIST) — 78.6% shared RTL
Mathematical & Protocol Scope	17 CCSDS Blue/Orange Books: $GF(2^8)$ Berlekamp dual-basis algebra, $GF(2^4)$ RS$(10, 6)$ AOS FHEC, Cycle-0 `FHP`, BCH$(63, 56)$, 4-phase Soft-OMS LDPC$(128, 64)$, $t = 16$ BM/Chien/Forney RS$(255, 223)$, 64-state Viterbi, 16-state Turbo, Two-Phase `FARM-1/SDLS AES-128-GCM` ($GF(2^{128})$ `GHASH`), 17-tap CSD 4-SPS SRRC, 3-stage Gardner STR, FLL-assisted Costas, Q32 S/X and X→Ka turnaround, and 6-component `T2B/T4B` PN ranging
S-Band RF Cascade Baseline	Lattice `LFMXO5-100T` + `ADRV9002` (`02_063978-02` clone) + `SKY66294-11` 4W PA → `RFSL2848` Isolator (`#3`) → `SYBD-20-272HP+` Coupler (`#2`) → Diplexer (`#4`: `D0421552` Path B at 2.51 W / `UPD215A` Path A 50% vacuum-derated to 1.41 W) → `SKY67153-396LF` 0.35 dB NF LNA
Silicon & Link Closure	**0 BRAM / 0 DSP**; 203.38–271.74 MHz Kintex UltraScale True $F_{\max}$; **34,560 / 34,560 RF Matrix Trials Passed**; +13.0 dB Downlink (2.0 Mbps Path B) / +9.4 dB (Derated Path A) and +36.6 dB Uplink (64 kbps) LEO link margins at 600 km ($\varepsilon = 10°$)

Division of Scope Between Volume I and Volume II: To avoid cross-volume duplication, **Volume I** serves as the practical hardware integration manual (pinout, register map, C11 HAL SDK reference, and LVDS/SPI board wiring). **Volume II (this monograph)** is dedicated strictly to the **first-principles mathematical derivations, finite-field proofs, DSP/FEC/cryptographic micro-architectures, and S-Band RF link/qualification physics**: **Part I (Chapter 1)** formulates the multi-clock queuing theory and coherent turnaround math; **Part II (Chapters 2–5)** derives the 16-bit unrolled CRC-16 equations, $GF(2^8)$ dual-basis bijections, Cycle-0 `FHP` induction proof, 4-phase Soft-OMS LDPC, 8-cycle Berlekamp-Massey/Forney RS decoder, and domain-separated `AES-128-GCM` security; **Part III (Chapters 6–7)** derives the physical 17-tap CSD SRRC filter, Gardner/Costas loops, CRT ranging, Friis noise cascade, 600 km LEO link budgets, and NASA/ECSS vacuum derating & radiation qualification; and **Part IV (Chapter 8)** analyzes the statistical E_b/N_0 waterfall curves and orbital pass dynamics.

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PART I

System Architecture & Multi-Clock Transponder Theory

1 Architectural Principles of Autonomous Spacecraft TT&C Transceivers

Unlike terrestrial wireless modems that optimize solely for peak spectral efficiency under continuous microprocessor supervision, an orbital spacecraft Telemetry, Tracking, and Command (TT&C) radio serves as the vehicle's non-negotiable survival lifeline. This opening chapter formulates the five physical and operational invariants that govern `CCSDS-SAT-TTC-IP` (`ccsds_sat_ttc_top v2.40.3`), proves why the dual-persona architecture eliminates flight-to-ground protocol divergence, and derives the mathematical coupling across the three closed-loop transponder feedback paths.

1.1 Five Foundational Invariants of Orbital Transceiver Design

Across Low-Earth Orbit (LEO), geostationary (GEO), cislunar, and deep-space missions, `ccsds_sat_ttc_top` enforces five architectural invariants:

1. **Zero Hard-Macro Vulnerability (`0 BRAM / 0 DSP`):** In radiation environments, FPGA Block RAMs (`RAMB36`) and DSP slices (`DSP48`) exhibit configuration and state Single-Event Functional Interrupts (SEFIs) that cannot be scrubbed without a full device reconfiguration. By realizing all multi-kilobyte CDC FIFOs in synchronous **Distributed LUTRAM (`RAM64M`, 1,200 LUTs)** guarded by continuous Hsiao $(39, 32)$ SECDED scrubbing and all FIR/NCO arithmetic in Canonical Signed-Digit (CSD) LUT adder trees, the entire flight persona consumes **0 Block RAMs, 0 DSP slices, and 13,864 Slice Flip-Flops**.
2. **Autonomous Command Recovery Without Software Execution:** During a solar proton event or OBC watchdog hang, the system bus clock (`clk_sys`) may stall. Clocking the entire uplink telecommand chain—from soft-LLR ASM synchronization and BCH$(63, 56)$/Short-LDPC$(128, 64)$ decoding down to the 3x TMR Emergency Hardware Command Pulse Decoder (`HCPD`, `VCID=63`)—from an independent 16 MHz crystal oscillator (`clk_tc_safe`) controlled by hardware mission straps (`ul_fec_override_en[1:0]`) guarantees commandability with zero CPU intervention.
3. **Multi-Protocol Unification:** Converges all five CCSDS Space Data Link protocols—`TC` [3], `TM` [2], `AOS` [5], `Proximity-1` [7], and `USLP` [6]—within a single datapath.
4. **Zero-Trust Cryptographic Atomicity:** Prevents the **`FARM-1/SDLS` Sequence-Lockout Paradox**—where an unauthenticated frame with a forged sequence number $N(S) = V(R)$ and valid CRC-16 advances $V(R)$ before SDLS MAC verification—by splitting `FARM-1` window checking (Phase A) and atomic $V(R)$ commit (Phase B) across a swipe-proof speculative rollback FIFO.
5. **Dual-Persona Mathematical Symmetry (`ROLE_MODE = 0, 1, 2`):** Unites the Spacecraft Flight transceiver (`ROLE_MODE = 0`, 23 modules), Hardware Ground EGSE (`ROLE_MODE = 1`, 22 modules), and Universal Crosslink/BIST transceiver (`ROLE_MODE = 2`, 28 modules) in a single parameterized hierarchy sharing 78.6% of the RTL modules.

1.2 Mathematical Formulation of the Three Coupled Transponder Loops

Referring to the top-level architecture and signal-flow diagrams in Volume I (**Figures 1.1–1.2**), three real-time feedback loops couple the uplink receiver to the downlink transmitter:

1. **Loop #1 — Atomic 4-Phase Toggle Handshake for Closed-Loop COP-1 (`FARM-1` → `CLCW` → `FOP-1`):** In the 16 MHz `clk_tc_safe` domain, `ccsds_farm1_tmr.sv` maintains the triplicated receiver sequence counter $V(R) \in [0..255]$ and formats the 32-bit `CLCW`. Because the downlink framer resides in `clk_sys` (100 MHz), the 32-bit `CLCW` word crosses via a 4-phase Request/Acknowledge toggle handshake (`clcw_req_tgl` / `clcw_ack_tgl` synchronized through 2-FF `ASYNC_REG` stages), guaranteeing that a multi-bit word is never sampled mid-transition.
2. **Loop #2 — Regenerative `T2B`/`T4B` Pseudo-Noise Ranging Turnaround:** Per CCSDS 414.1-B-3 [14], the composite ranging sequence of period $L = 2 \times 7 \times 11 \times 15 \times 19 \times 23 = 1{,}009{,}470$ chips is acquired in `ccsds_pn_ranging.sv` and regenerated noise-free onto the downlink quadrature channel, eliminating the thermal-noise turnaround loss of analog bent-pipe transponders.

3. **Loop #3 — Fixed-Point Q32 Coherent Turnaround Frequency Synthesis (S, X, X→Ka, and Ka→Ka):** Per CCSDS 401.0-B-32 [13] (Tables 2.4-1 and 2.5-1), two-way coherent Doppler tracking requires multiplying the tracked uplink carrier frequency f_{ul} by an exact rational turnaround ratio M/N to synthesize the downlink carrier $f_{\text{dl}} = (M/N)f_{\text{ul}}$. It is essential to distinguish between **intra-band** turnaround ratios and **X-to-Ka cross-band** turnaround ratios:

   - *S-Band Uplink (*2025–$2120\,$*MHz)* $\rightarrow$ *S-Band Downlink (*2200–$2300\,$*MHz):* $M/N = 240/221 \approx 1.085973$.
   - *X-Band Uplink (*7145–$7235\,$*MHz)* $\rightarrow$ *X-Band Downlink (*8400–$8500\,$*MHz):* $M/N = 880/749 \approx 1.174900$.
   - *X-Band Uplink (*7145–$7235\,$*MHz)* $\rightarrow$ *Ka-Band Downlink (*31.8–$32.3\,$*GHz) Cross-Band:* $M/N = 3344/749 \approx 4.464619$ (Category B Deep Space) and $M/N = 3360/749 \approx 4.485981$ or $3328/749 \approx 4.443258$ (Category A Near-Earth).
   - *Ka-Band Uplink (*34.2–$34.7\,$*GHz)* $\rightarrow$ *Ka-Band Downlink (*31.8–$32.3\,$*GHz) Intra-Band:* $M/N = 3344/3599 \approx 0.929147$ (Category B) or $3328/3599 \approx 0.924701$ (Category A).

   In `rtl/ccsds_phy_modem.sv` (lines 380–386), the tracked 32-bit uplink NCO Frequency Control Word (FCW_{rx}) is scaled without combinational division using pre-computed Q32 fixed-point reciprocal multipliers $K_{\text{Q32}}(M/N) = \lfloor (M/N) \cdot 2^{32} + 0.5 \rfloor$:

$$\begin{aligned} \text{FCW}_{\text{tx}} &= (\text{FCW}_{\text{rx}} \times K_{\text{Q32}}(M/N)) \gg 32, \\ K_{\text{Q32}}(240/221) &= \texttt{64'h0000_0001_1602_5117}, \qquad K_{\text{Q32}}(880/749) = \texttt{64'h0000_0001_2CC6_7C21}, \\ K_{\text{Q32}}(3344/749) &= \texttt{64'h0000_0004_76F0_D8FC}, \qquad K_{\text{Q32}}(3344/3599) = \texttt{64'h0000_0000_EDDD_718B}, \end{aligned} \tag{1.1}$$

   achieving $< 2.3 \times 10^{-10}$ relative frequency accuracy in a single pipelined register stage.

PART II

Algebraic Foundations, Data Link Framing, Forward Error Correction & Zero-Trust Security

2 Finite-Field Algebra, Spectral Randomization & Fault-Tolerant Memory

Before a space link can frame packets or correct channel errors, it requires a rigorous foundation in finite-field algebra ($\mathbb{F}_2[x]$, $GF(2^4)$, and $GF(2^8)$) and radiation-tolerant storage. This chapter derives the complete 16-bit unrolled parallel transition equations for CRC-16-CCITT, formulates the additive pseudo-randomizers and Berlekamp dual-basis bijection, derives the shortened $GF(2^4)$ RS$(10, 6)$ AOS header protector, and proves how our Hsiao $(39, 32)$ SECDED Distributed-LUTRAM FIFOs guarantee swipe-proof speculative rollback.

2.1 Parallel Companion Matrix Derivation for CRC-16 and CRC-32

Frame error detection across all five CCSDS Space Data Link protocols relies on polynomial division over $\mathbb{F}_2[x]$ implemented in `rtl/ccsds_crc_universal.sv`:

$$G_{16}(x) = x^{16} + x^{12} + x^5 + 1 \quad (\texttt{16'h1021}, \text{ seed } \texttt{16'hFFFF}, \text{ valid residue } \texttt{16'h0000 / 16'h1D0F}), \tag{2.1}$$

$$G_{32}(x) = x^{32} + x^{23} + x^{21} + x^{11} + x^2 + 1 \quad (\texttt{32'h00A0_0805}, \text{ seed } \texttt{32'hFFFF_FFFF}). \tag{2.2}$$

Let $c_{\text{in}}[15:0]$ denote the current 16-bit CRC-16 register and $d[7:0]$ denote the incoming AXI4-Stream byte (processed MSB-first from $b = 7$ down to $b = 0$ as in `step_crc16_byte`). Define the 8-bit upper XOR vector $x_i = c_{\text{in}}[8+i] \oplus d[i]$ for $i \in [0..7]$. Why do the lower 8 incoming state bits $c_{\text{in}}[7:0]$ appear only in $c_{\text{out}}[15:8]$ and never in $c_{\text{out}}[7:0]$? Because each of the 8 single-bit left shifts moves $c_{\text{in}}[7:0]$ up by 8 bit positions into $c_{\text{out}}[15:8]$ without ever reaching the MSB feedback tap $c[15]$ during that byte! Only $c_{\text{in}}[15:8] \oplus d[7:0] = x[7:0]$ crosses the feedback tap during the 8 shifts, with the x^{12} tap coupling $x_7..x_4$ back into $x_3..x_0$. Unrolling all 8 shifts yields the complete, exact 16-bit

closed-form XOR transition equations:

$$\begin{aligned}
c_{\text{out}}[0] &= x_0 \oplus x_4, & c_{\text{out}}[8] &= c_{\text{in}}[0] \oplus x_3 \oplus x_7,\\
c_{\text{out}}[1] &= x_1 \oplus x_5, & c_{\text{out}}[9] &= c_{\text{in}}[1] \oplus x_4,\\
c_{\text{out}}[2] &= x_2 \oplus x_6, & c_{\text{out}}[10] &= c_{\text{in}}[2] \oplus x_5,\\
c_{\text{out}}[3] &= x_3 \oplus x_7, & c_{\text{out}}[11] &= c_{\text{in}}[3] \oplus x_6,\\
c_{\text{out}}[4] &= x_4, & c_{\text{out}}[12] &= c_{\text{in}}[4] \oplus x_0 \oplus x_4 \oplus x_7,\\
c_{\text{out}}[5] &= x_0 \oplus x_4 \oplus x_5, & c_{\text{out}}[13] &= c_{\text{in}}[5] \oplus x_1 \oplus x_5,\\
c_{\text{out}}[6] &= x_1 \oplus x_5 \oplus x_6, & c_{\text{out}}[14] &= c_{\text{in}}[6] \oplus x_2 \oplus x_6,\\
c_{\text{out}}[7] &= x_2 \oplus x_6 \oplus x_7, & c_{\text{out}}[15] &= c_{\text{in}}[7] \oplus x_3 \oplus x_7.
\end{aligned} \tag{2.3}$$

Every bit of $c_{\text{out}}[15:0]$ is a function of at most 4 XOR inputs, evaluating in a **single 6-input LUT delay** per byte and producing the exact CCSDS standard check value CRC16(`"123456789"`) = `16'h29B1`.

2.2 Telemetry and Telecommand Pseudo-Randomizers

To guarantee sufficient symbol-transition density for Gardner timing recovery and disperse periodic spectral lines, `rtl/ccsds_lfsr_scrambler.sv` implements both CCSDS additive (synchronous) pseudo-randomizers over period $2^8 - 1 = 255$ bits ([8, 9]):

$$h_{\text{TM}}(x) = x^8 + x^7 + x^5 + x^3 + 1 \quad (\text{tap mask } \texttt{8'hA9},\ \text{init } \texttt{8'hFF} \implies \texttt{FF 48 0E C0 9A}\ldots), \tag{2.4}$$

$$h_{\text{TC}}(x) = x^8 + x^6 + x^4 + x^3 + x^2 + x + 1 \quad (\text{tap mask } \texttt{8'h5F},\ \text{init } \texttt{8'hFF} \implies \texttt{FF 39 9E 5A 68}\ldots). \tag{2.5}$$

Both generators initialize to all-ones (`8'hFF`) immediately following the unscrambled Attached Sync Marker (ASM) and step 8 LFSR states per clock cycle to XOR-scramble or de-scramble one full byte per cycle.

2.3 Galois Field $GF(2^8)$ & Berlekamp Dual-Basis Bijection

Per CCSDS 131.0-B-5 [8], downlink Reed-Solomon RS$(255, 223)$ codewords are constructed over the finite field $GF(2^8) \cong \mathbb{F}_2[x]/\langle F(x)\rangle$ defined by the primitive polynomial:

$$F(x) = x^8 + x^7 + x^2 + x + 1 \quad (\texttt{9'h187}), \qquad F(\alpha) = 0 \implies \alpha^8 = \alpha^7 + \alpha^2 + \alpha + 1. \tag{2.6}$$

While internal syndrome, Berlekamp-Massey, Chien, and Forney arithmetic is naturally expressed in the **standard polynomial basis** $\{\alpha^7, \alpha^6, \ldots, \alpha^1, 1\}$ where $u \in GF(2^8)$ has coordinate vector $\mathbf{u} = [u_7, u_6, \ldots, u_0]^T$, CCSDS 131.0-B-5 Annex B mandates that symbols on the space link be represented in **Berlekamp's dual basis** $\{z_0, z_1, \ldots, z_7\}$ orthogonal to $\{\ell_0, \ldots, \ell_7\} = \{1, \alpha, \ldots, \alpha^7\}$ under the trace bilinear form $\text{Tr}(\gamma\beta_i\ell_j) = \delta_{ij}$. In `rtl/ccsds_pkg.sv` and `rtl/ccsds_gf256_math.sv`, the change-of-basis transformation $\mathbf{z} = T_\alpha \mathbf{u} \pmod 2$ and its exact inverse $\mathbf{u} = T_\alpha^{-1}\mathbf{z} \pmod 2$ ($T_\alpha T_\alpha^{-1} = I_8$ over $\mathbb{F}_2$) are realized as zero-latency 8-bit XOR matrix-vector products:

$$\begin{aligned}
T_\alpha &= \big[\texttt{0xFE, 0x69, 0x6B, 0x0D, 0xEF, 0xF2, 0x5B, 0xC7}\big]^T \quad (\alpha^0 \mapsto \texttt{0x7B},\ \alpha^7 \mapsto \texttt{0x8D}),\\
T_\alpha^{-1} &= \big[\texttt{0x9B, 0xDD, 0x3E, 0x1C, 0x66, 0x33, 0xA9, 0x2D}\big]^T.
\end{aligned} \tag{2.7}$$

2.4 Shortened RS$(10, 6)$ over $GF(2^4)$ for AOS Header Protection

In CCSDS 732.0-B-4 AOS frames [5], the 24 protected header bits (`TFVN`, `SCID`, `VCID`, `Replay`, `Usage`, `Rsvd`, and `VC_Cycle`) are partitioned into six 4-bit nibbles $m_5, m_4, \ldots, m_0 \in GF(2^4)$ modulo $p(x) = x^4 + x + 1$ (`5'h13`) and encoded with a shortened RS$(10, 6)$ code of minimum distance $d_{\min} = 5$:

$$g_{\text{FHEC}}(x) = \prod_{j=6}^{9}(x + \alpha^j) = x^4 + \alpha^3 x^3 + \alpha^1 x^2 + \alpha^3 x + 1 \implies (g_3, g_2, g_1, g_0) = (\texttt{4'h8, 4'h2, 4'h8, 4'h1}). \tag{2.8}$$

On receive, let $S_j = R(\alpha^{6+j}) \in GF(2^4)$ for $j \in \{0, 1, 2, 3\}$ be the four 4-bit syndromes of the 10-nibble header. If a single nibble at index $k \in [0..9]$ is corrupted by error pattern $e \in GF(2^4) \setminus \{0\}$, then $S_0 = e\alpha^{6k}$ and $S_1 = e\alpha^{7k}$, so the ratio $S_1/S_0 = \alpha^k$ uniquely identifies the corrupted nibble position k while $e = S_0\alpha^{-6k}$ gives the exact 4-bit error mask—correcting up to 4 consecutive bit flips in a nibble in a single clock cycle.

2.5 Odd-Weight Hsiao $(39, 32)$ SECDED LUTRAM FIFOs & Speculative Rollback

In `rtl/ccsds_cdc_fifo_ecc.sv` and the 64-entry SDLS Security Association RAM, every 32-bit word $D[31:0]$ is encoded with a 7-bit **Hsiao** $(39, 32)$ **SECDED** parity vector $P[6:0]$ constructed from odd-weight parity masks:

$$\begin{aligned} H_{\text{Hsiao}} &= \big[\texttt{32'h56AA_AD5B},\ \texttt{32'h9B33_366D},\ \texttt{32'hE3C3_C78E},\\ &\qquad \texttt{32'h03FC_07F0},\ \texttt{32'h03FF_F800},\ \texttt{32'hFC00_0000}\big]^T,\\ P[5:0] &= H_{\text{Hsiao}}\, D[31:0] \pmod 2, \qquad P[6] = \bigoplus_{i=0}^{31} D[i] \oplus \bigoplus_{j=0}^{5} P[j]. \end{aligned} \tag{2.9}$$

Because the storage array `mem_ecc_reg` is written inside a dedicated synchronous process without asynchronous reset sensitivity, synthesis maps all FIFO storage into **Distributed LUTRAMs (`RAM64M`, 1,200 LUTs across the chip)** rather than discrete Flip-Flops. To prevent unauthenticated uplink telecommand bytes from leaking across the `clk_tc_safe` $\rightarrow$ `clk_sys` boundary, `ccsds_cdc_fifo_ecc.sv` maintains three write-domain pointers: the committed boundary ptr_{comm}, the active speculative write pointer ptr_{spec}, and the end-of-frame snapshot ptr_{eof} (`frame_end_ptr_q`). When Frame N passes CRC-16, $\text{ptr}_{\text{eof}} \leftarrow \text{ptr}_{\text{spec}}$ is latched immediately—even if Frame $N+1$ begins arriving in the same CLTU while Frame N is still undergoing SDLS MAC verification. Asserting `commit_pulse` updates $\text{ptr}_{\text{comm}} \leftarrow \text{ptr}_{\text{eof}}$, exposing only authenticated Frame N to the read port, whereas `rollback_pulse` rewinds $\text{ptr}_{\text{spec}} \leftarrow \text{ptr}_{\text{comm}}$.

3 Unified Space Data Link Layer: Cycle-0 `FHP` & Multi-Frame CLTU Theory

While Volume I, Chapters 3–4 catalog the packed header structures and bitfields for `TM`, `TC`, `AOS`, `Proximity-1`, and `USLP`, this chapter derives the two core algorithmic innovations in the framing layer: the **Cycle-0 closed-form `First_Hdr_Ptr` (`FHP`) recurrence** and the **multi-frame CLTU `0x55` fill-stripping automaton**.

3.1 Induction Proof of Cycle-0 Closed-Form `First_Hdr_Ptr` (`FHP`) Calculation

In continuous downlink protocols (`TM`, `AOS`, and `USLP`), variable-length Space Packets are multiplexed back-to-back across fixed-length Transfer Frame Data Fields (`TFDF`) of capacity $L_{\text{TFDF}} \in [1..1024]$ bytes. Because the 48-bit TM Primary Header is transmitted *before* the payload zone, the framer must determine on **Cycle 0** the exact byte offset $\texttt{FHP} \in [0..L_{\text{TFDF}} - 1]$ where the first new Space Packet header begins inside that frame.

Rather than buffering an entire frame in Block RAM to scan for packet headers retroactively, `rtl/ccsds_vc_arbiter_mux.sv` exploits the fact that every CCSDS 133.0-B-2 Space Packet header declares its total length $L_{\text{pkt}} = 6 + (\texttt{pkt_data_len} + 1)$ in bytes $4..5$. Let $R_k \geq 0$ (`rem_in_cur_pkt_q`) denote the number of residual un-transmitted bytes of an in-progress Space Packet at the start of Frame k (with $R_0 = 0$ at reset). By induction on k:

$$\texttt{FHP}_k = \begin{cases} \texttt{11'h7FE} & \text{if } R_k == 0 \text{ and no Virtual Channel has a valid packet ready (Only-Idle-Data frame)},\\ \texttt{11'h000} & \text{if } R_k == 0 \text{ and a selected VC starts a new Space Packet at Byte 0},\\ R_k[10:0] & \text{if } 0 < R_k < L_{\text{TFDF}} \text{ (spillover ends at byte } R_k - 1\text{; next packet starts at offset } R_k),\\ \texttt{11'h7FF} & \text{if } R_k \geq L_{\text{TFDF}} \text{ (jumbo packet spans entire frame; } R_{k+1} = R_k - L_{\text{TFDF}}). \end{cases} \tag{3.1}$$

Within Frame k, whenever a new Space Packet header is ingested at byte index $b \in [0..L_{\text{TFDF}} - 1]$ with declared length L_{pkt}, the residual counter updates via $R_{\text{next}} = \max(0,\ b + L_{\text{pkt}} - L_{\text{TFDF}})$. If a packet finishes at $b + L_{\text{pkt}} < L_{\text{TFDF}}$ and the active VC queue goes empty, the arbiter immediately injects deterministic CCSDS Idle Packets (`APID = 11'h7FF`) to fill the remainder of L_{TFDF}, guaranteeing that the next packet header starts at the exact byte offset R_k predicted on Cycle 0 with **zero frame-buffering latency and zero Block RAM**.

3.2 Multi-Frame CLTU Delimitation & `0x55` Fill-Octet Stripping Automaton

Under CCSDS 231.0-B-4 [9], an uplink Command Link Transmission Unit (`CLTU`) encapsulates one or more variable-length TC or USLP Transfer Frames inside a sequence of 7-octet (56-bit) BCH$(63, 56)$ information blocks terminated by the 64-bit Tail Sequence (`64'hC5C5_C5C5_C5C5_C579`). When the total Transfer Frame length $L_{\text{exact}} = \texttt{frame_len} + 1$ is not an integer multiple of 7 octets, the ground encoder (`ccsds_bch6356_enc.sv`) pads the final codeblock with $p = (7 - (L_{\text{exact}} \bmod 7)) \bmod 7 \in [0..6]$ alternating `8'h55` (01010101_2) fill octets.

How does the spacecraft deframer (`ccsds_tc_uslp_deframer.sv`) distinguish between trailing `0x55` padding octets and a second Transfer Frame concatenated back-to-back inside the same CLTU?

1. During the first 5 octets of a frame (ST_HDR), the deframer extracts the 10-bit TC frame_len field (or 16-bit USLP length field) and loads a down-counter $C_{\text{rem}} \leftarrow L_{\text{exact}} - 5$.
2. On the exact clock cycle where C_{rem} reaches 0 (ST_CRC_CHECK), the deframer evaluates the 16-bit CRC-16 residue $R_{\text{CRC}} == 16'h0000$ and emits tlast = 1'b1 to the speculative rollback FIFO.
3. In the following state (ST_SKIP_FILL), if the next decoded octet equals 8'h55 (or the CLTU tail pulse asserts), the octet is silently discarded as padding; if the octet is non-8'h55 (since a valid TC Primary Header starts with TFVN = 2'b00 $\implies$ Byte $0 \in [0x00..0x3F] \neq 0x55$), the FSM transitions directly back to ST_HDR to parse the concatenated frame without losing a single byte.

4 Forward Error Correction: Algebraic Block Codes to Iterative Soft-Decision Decoders

This chapter derives the mathematical formulations of the five Forward Error Correction (FEC) engines in ccsds_sat_ttc_top, progressing from phase-ambiguous soft-LLR synchronization and algebraic BCH$(63, 56)$ syndrome decoding to our 4-phase time-multiplexed Short-LDPC$(128, 64)$ Soft-OMS iterative decoder and 8-cycle unrolled $t = 16$ Berlekamp-Massey RS$(255, 223)$ decoder.

4.1 Differential Soft-LLR ASM Correlation Under $0°/180°$ Phase Ambiguity

Because suppressed-carrier BPSK Costas loops exhibit a two-fold ($0°/180°$) phase lock ambiguity, the demodulated 4-bit signed Log-Likelihood Ratio (LLR) stream $\ell_k \in [-8, +7]$ may arrive inverted. In rtl/ccsds_asm_correlator.sv, a sliding window of $M \in \{16, 24, 32, 64\}$ soft LLRs is correlated simultaneously against both the true Attached Sync Marker pattern $p_i \in \{-1, +1\}$ (16'hEB90 for CLTU, 32'h1ACFFC1D for TM/AOS, 24'hFAF320 for Proximity-1, and 64'h034776C7272895B0 for Short-LDPC/AR4JA) and its bitwise complement:

$$\Gamma_{\text{corr}} = \sum_{i=0}^{M-1} p_i \, \text{sgn}(\ell_{k-i}), \qquad \text{Lock} \iff |\Gamma_{\text{corr}}| \geq \Gamma_{\text{thresh}}, \qquad \text{Invert} \iff \Gamma_{\text{corr}} \leq -\Gamma_{\text{thresh}}. \tag{4.1}$$

When inverted lock is detected, the correlator automatically negates all subsequent LLRs ($\ell_k \leftarrow -\ell_k$) across the frame.

4.2 Uplink Expurgated BCH$(63, 56)$ SEC-DED Codec

Per CCSDS 231.0-B-4 [9], each 64-bit uplink CLTU codeblock contains 56 information bits $m(x)$, 7 parity bits $r(x) = \overline{(x^7 m(x) \bmod g_{\text{BCH}}(x))}$ generated by the expurgated polynomial $g_{\text{BCH}}(x) = (x^6 + x + 1)(x + 1) = x^7 + x^6 + x^2 + 1$ (7'h45), and 1 trailing filler bit (0). In rtl/ccsds_bch6356_dec.sv, the 7-bit syndrome $s(x) = c_{62..0}(x) \bmod g_{\text{BCH}}(x)$ is decomposed into the 6-bit primitive syndrome $s_6 \in GF(2^6)$ and overall codeblock parity $p_{\text{odd}} = s(1)$:

- $s_6 == 0 \wedge p_{\text{odd}} == 0$: Codeblock is error-free; all 7 de-randomized information octets are emitted.
- $s_6 \neq 0 \wedge p_{\text{odd}} == 1$: Single-bit error detected; a 63-entry discrete-logarithm lookup table maps $s_6 = \alpha^k$ to the exact corrupted bit index $k \in [0..62]$, healing the bit in hardware and incrementing cnt_secded_corr_q.
- $s_6 \neq 0 \wedge p_{\text{odd}} == 0$: Uncorrectable double-bit error or intentional CLTU Tail Sequence (64'hC5C5_C5C5_C5C5_C579, which produces the deterministic tail syndrome 7'h25), cleanly terminating CLTU reception.

4.3 Uplink 4-Phase Time-Multiplexed Short-LDPC$(128, 64)$ Soft-OMS Decoder

For low-SNR safe-mode links and deep-space telecommanding, CCSDS 231.1-O-1 [10] specifies a rate-$1/2$ $(128, 64)$ quasi-cyclic Low-Density Parity-Check (LDPC) code built from 16×16 circulant submatrices I_s (the 16×16 identity matrix circularly shifted right by s columns) and $\mathbf{0}$:

$$H_{64\times128} = \begin{bmatrix} I_0 \oplus I_7 & I_2 & I_{14} & I_6 & \mathbf{0} & I_0 & I_{13} & I_0 \\ I_6 & I_0 \oplus I_{15} & I_0 & I_1 & I_0 & \mathbf{0} & I_0 & I_7 \\ I_4 & I_1 & I_0 \oplus I_{15} & I_{14} & I_{11} & I_0 & \mathbf{0} & I_3 \\ I_0 & I_1 & I_9 & I_0 \oplus I_{13} & I_{14} & I_1 & I_0 & \mathbf{0} \end{bmatrix}, \qquad G_{64\times128} = \left[\, I_{64} \;\middle|\; W_{64\times64} \,\right]. \tag{4.2}$$

In `rtl/ccsds_ul_short_ldpc_dec.sv`, we fold each decoding iteration across **4 time-multiplexed phases (16 check rows per clock cycle)** using the **Scaled Offset Min-Sum (Soft-OMS)** algorithm. For check node m and neighboring variable nodes $n \in \mathcal{N}(m)$, the check-to-variable extrinsic message $L_{m\to n}^{(t)}$ and updated variable posterior $Q_n^{(t)}$ are:

$$L_{m\to n}^{(t)} = \left(\prod_{n' \in \mathcal{N}(m)\setminus\{n\}} \operatorname{sgn}\big(q_{n'\to m}^{(t-1)}\big) \right) \cdot \max\left(\min_{n' \in \mathcal{N}(m)\setminus\{n\}} \big|q_{n'\to m}^{(t-1)}\big| - \beta_{\text{oms}},\ 0 \right), \tag{4.3}$$

$$Q_n^{(t)} = \ell_n^{\text{ch}} + \sum_{m \in \mathcal{M}(n)} L_{m\to n}^{(t)}, \qquad \text{Early Stop} \iff \mathbf{s}^{(t)} = \hat{\mathbf{c}}^{(t)} H^T = \mathbf{0} \pmod 2. \tag{4.4}$$

This 4-phase folded pipeline reduces the combinational critical path to just **11 logic levels (**110.22 **MHz True** $F_{\max}$ **on Zynq-7010,** 203.38 **MHz on Kintex UltraScale)** while achieving a $+6.3$ dB coding gain over uncoded BPSK at BER $= 10^{-6}$.

4.4 Downlink Dual-Basis RS$(255, 223)$ Codec ($t = 16$ Berlekamp-Massey, Chien & Forney)

The downlink Reed-Solomon outer codec (`ccsds_dl_rs255_223_enc.sv` and `ccsds_dl_rs255_223_dec.sv`) operates over the symmetric 32-root generator polynomial mandated by CCSDS 131.0-B-5 [8]:

$$g_{\text{RS}}(x) = \prod_{j=112}^{143} (x - \alpha^{11j}) = \sum_{k=0}^{32} g_k x^k, \qquad \alpha^{11\times 112} = \alpha^{212}, \ldots, \alpha^{11\times 143} = \alpha^{43}. \tag{4.5}$$

In the Ground EGSE persona (`ROLE_MODE = 1/2`), `ccsds_dl_rs255_223_dec.sv` maps each incoming dual-basis byte into standard basis via T_α^{-1} and executes a 4-stage hardware algebraic decoding pipeline capable of correcting any $t \le 16$ symbol errors per 255-byte codeword:

1. **Horner Syndrome Evaluation (`ST_ACCUM`):** Evaluates all 32 syndromes $S_k = R(\alpha^{11(112+k)})$ for $k \in [0..31]$ on the fly as bytes stream in ($S_k \leftarrow (S_k \otimes \alpha^{11(112+k)}) \oplus r_i$). If $\bigvee_{k=0}^{31} S_k == 0$, the codeword is emitted with zero correction latency.
2. **8-Cycle Unrolled Berlekamp-Massey Solver (`RS_DEC_SOLVE_BM`):** Solves the key equation $\Lambda(z)S(z) \equiv \Omega(z) \pmod{z^{32}}$ for the degree-ν ($\nu \le 16$) error-locator polynomial $\Lambda(z) = 1 + \Lambda_1 z + \cdots + \Lambda_\nu z^\nu$ in just 8 clock cycles (4 Berlekamp-Massey discrepancy updates $d_n = S_n \oplus \bigoplus_{i=1}^{L} \Lambda_i \otimes S_{n-i}$ per clock).
3. **Forney Error-Evaluator Polynomial (`RS_DEC_SOLVE_OMEGA`):** Computes the degree-15 evaluator polynomial $\Omega(z) = S(z)\Lambda(z) \bmod z^{32}$ in 2 clock cycles.
4. **Pipelined Chien Search & Forney Correction (`DEC_CHIEN / DEC_FORNEY`):** Sweeps $x_i = \alpha^{-(254-i)}$ across the buffered codeword. Whenever $\Lambda(x_i^{-1}) = \Lambda_{\text{even}}(x_i^{-1}) \oplus \Lambda_{\text{odd}}(x_i^{-1}) == 0$ (where $\Lambda_{\text{odd}}(x_i^{-1}) = x_i^{-1}\Lambda'(x_i^{-1})$ in characteristic 2), the exact error magnitude e_i is computed via Forney's formula with root-offset compensation $\alpha^{11\times 112} = \alpha^{212}$ and XOR-healed back into dual basis:

$$e_i = \frac{x_i^{112\times 11}\,\Omega(x_i^{-1})}{\Lambda'(x_i^{-1})} = \frac{x_i^{212}\,\Omega(x_i^{-1})}{x_i\,\Lambda_{\text{odd}}(x_i^{-1})}, \qquad \hat{c}_{\text{dual},i} = r_{\text{dual},i} \oplus T_\alpha(e_i). \tag{4.6}$$

5 Zero-Trust Space Data Link Security (SDLS), COP-1 ARQ & Radiation Mitigation

Having corrected channel bit errors in Chapter 4, the receiver must verify cryptographic authenticity and frame sequence order without exposing the spacecraft to Denial-of-Service lockouts or Galois/Counter Mode (GCM) Initialization Vector (IV) reuse. This chapter formulates the **Two-Phase `FARM-1/SDLS` Speculative Commit/Rollback Protocol**, derives the **domain-separated 96-bit `AES-128-GCM` IV and 4-bit/clock** $GF(2^{128})$ **`GHASH` engine**, and analyzes the 3x Triple-Modular Redundancy (TMR) voters.

5.1 Two-Phase FARM-1 & SDLS Speculative Commit Proof

Let $V(R) \in [0..255]$ denote the committed receiver sequence counter in `ccsds_farm1_tmr.sv` and let $\text{Seq}_{\text{SA}}[v] \in [0..2^{32}-1]$ denote the monotonic anti-replay counter for Virtual Channel $v \in [0..63]$ in `ccsds_sdls_engine.sv`. Rather than advancing $V(R)$ before authentication, the receiver executes a two-phase transition across `clk_tc_safe` and `clk_sys`:

1. **Phase A (`clk_tc_safe`):** `ccsds_farm1_tmr.sv` verifies $N(S) == V(R)$ while holding $V(R)$ frozen, and streams the frame into the speculative rollback FIFO (`u_rx_commit_cdc_fifo`).
2. **Phase B (`clk_sys`):** `ccsds_sdls_engine.sv` verifies $\text{Seq}_{\text{rx}} > \text{Seq}_{\text{SA}}[v]$ and $T_{\text{calc}} == T_{\text{rx}}$ before asserting either `sdls_commit_pulse` or `sdls_rollback_pulse`.

This decoupled transition guarantees two formal security invariants:

1. **Spoof Immunity ($T_{\text{calc}} \neq T_{\text{rx}}$ or $\text{Seq}_{\text{rx}} \leq \text{Seq}_{\text{SA}}[v]$):** `sdls_rollback_pulse` rewinds the speculative FIFO write pointer and leaves both $V(R)$ and $\text{Seq}_{\text{SA}}[v]$ invariant. An adversary injecting arbitrary CRC-valid frames cannot advance $V(R)$ or trigger `FARM-1` Lockout.
2. **Out-of-Sequence Rejection ($N(S) \neq V(R)$):** Frames outside the current expectation are rejected in Phase A before reaching `ccsds_sdls_engine.sv`, ensuring that dropped frames or Go-Back-N retransmissions never prematurely advance $\text{Seq}_{\text{SA}}[v]$.

5.2 Domain-Separated AES-128-GCM IV & $GF(2^{128})$ GHASH

In NIST SP 800-38D Galois/Counter Mode (`AES-128-GCM`), encrypting two distinct frames under the same 128-bit key K and identical 96-bit Initialization Vector (IV_{96}) leaks the XOR of their plaintexts and allows polynomial root recovery of the hash subkey $H = \text{AES}_K(0^{128})$ (the GCM "Forbidden Attack").

To guarantee global IV_{96} uniqueness across all 64 Virtual Channels and across multiple spacecraft sharing a mission key, `rtl/ccsds_sdls_engine.sv` (line 206, `format_aes_ctr_block`) constructs the 128-bit counter block $J_j = \{\text{IV}_{96},\ \texttt{16'h0000},\ \text{blk_idx}[15:0]\}$ using the 16-bit Security Parameter Index ($\text{SPI}[15:0]$) and 32-bit monotonic sequence number ($\text{Seq}[31:0]$) from the 6-byte CCSDS 355.0-B-2 Security Header, framed by fixed domain constants (`16'hCC5D` and `32'h5344_4C53` = ASCII `"SDLS"`):

$$\begin{aligned} \text{IV}_{96} &= \big\{\ \texttt{16'hCC5D},\ \underbrace{\text{SCID}[9:0],\ \text{VCID}[5:0]}_{\text{Domain-Separated SPI}[15:0]},\ \text{Seq}[31:0],\ \texttt{32'h5344_4C53}\ \big\}, \\ J_0 &= \big\{\ \text{IV}_{96},\ \texttt{32'h0000_0001}\ \big\}, \qquad J_j = \big\{\ \text{IV}_{96},\ \texttt{16'h0000},\ (j+1)[15:0]\ \big\}. \end{aligned} \tag{5.1}$$

Because each Virtual Channel $v \in [0..63]$ on Spacecraft SCID $\in [0..1023]$ is provisioned in the 64-entry SA RAM with its unique domain-separated $\text{SPI} = \{\text{SCID}[9:0], \text{VCID}[5:0]\}$ (see Volume I, Section 6.3), the tuple $(\text{SCID}, \text{VCID}, \text{Seq})$ is injectively embedded inside IV_{96}, making cross-VC or cross-spacecraft IV collisions mathematically impossible.

To avoid instantiating separate AES cores for key expansion, counter-stream encryption, and hash-subkey derivation, `ccsds_sdls_engine.sv` time-multiplexes a single 10-round AES-128 datapath across three phases per frame:

1. **Phase 1 (Hash Subkey H):** Evaluates $H = \text{AES}_K(0^{128})$ in 10 clock cycles.
2. **Phase 2 (Initial Counter Mask $E_K(J_0)$):** Evaluates $E_K(J_0)$ in 10 clock cycles and feeds the 48-bit Security Header $\{\text{SPI}[15:0], \text{Seq}[31:0]\}$ padded with 0^{80} into the `GHASH` accumulator as Additional Authenticated Data (A).
3. **Phase 3 (Payload Counter Stream & 4-Bit/Clock $GF(2^{128})$ GHASH):** For each 128-bit payload block $j \geq 1$, XORs $E_K(J_j)$ with the plaintext/ciphertext and updates $Y_j = (Y_{j-1} \oplus C_j) \cdot H$ over $GF(2^{128}) \cong \mathbb{F}_2[x]/\langle 1 + x + x^2 + x^7 + x^{128} \rangle$ (`ghash_step_4bit`, 4 bits per clock = 32 clocks per 128-bit block, requiring **1 LUT6 logic level and 0 DSP slices**). After the length block $\text{len}(A)_{64} \| \text{len}(C)_{64}$, the 128-bit authentication tag is $T = Y_m \oplus E_K(J_0)$.

5.3 Triple-Modular Redundancy (3x TMR) & Single-Cycle Auto-Scrubbing

Every state register in `ccsds_farm1_tmr.sv`, `ccsds_hcpd_ctrl.sv`, and `ccsds_axi_lite_csr.sv` is triplicated across three preserved flip-flop banks (`*_a_q`, `*_b_q`, `*_c_q` guarded by `(* syn_preserve = 1, dont_touch = "true" *)`) and voted combinationally on every cycle:

$$v = \text{vote3}(a,b,c) = (a \wedge b) \vee (b \wedge c) \vee (a \wedge c), \qquad e_{\text{seu}} = (a \oplus b) \vee (b \oplus c). \tag{5.2}$$

Whenever an ionizing particle flips a bit in any replica ($e_{\text{seu}} \neq 0$), vote3 masks the fault with zero output latency, writes the clean voted state v back into all three replicas on the next clock edge, increments `cnt_tmr_seu_corr_q` (`0x18[31:16]`), and pulses `edac_scrub_req_out`.

PART III

All-Digital Baseband RF Modem DSP & S-Band Transceiver Hardware Engineering

6 All-Digital Baseband RF Modem DSP & Regenerative Pseudo-Noise Ranging

Moving from the protocol and security planes down to the physical waveform boundary, `rtl/ccsds_phy_modem.sv` and `rtl/ccsds_pn_ranging.sv` synthesize and demodulate complex baseband I/Q sample streams in the 120 MHz `clk_adc_iq` domain. This chapter derives the exact 17-tap Canonical Signed-Digit (CSD) Square-Root Raised-Cosine (SRRC) filter implemented in RTL, the 3-stage pipelined Gardner symbol timing loop, the FLL-assisted Costas carrier recovery loop, and the 6-component regenerative PN ranging correlator.

6.1 Multiplier-Free CSD 17-Tap 4-SPS SRRC Filter ($\alpha = 0.35$)

Per CCSDS 401.0-B-32 [13], baseband pulse shaping uses a Square-Root Raised-Cosine (SRRC) transfer function with roll-off $\alpha = 0.35$ across $t \in [-2T_s, +2T_s]$ at $N_{\text{sps}} = 4$ Samples Per Symbol (17 taps $n \in [0..16]$, center peak $n = 8$):

$$h_{\text{SRRC}}(t) = \frac{\sin\left[\pi \frac{t}{T_s}(1-\alpha)\right] + 4\alpha \frac{t}{T_s} \cos\left[\pi \frac{t}{T_s}(1+\alpha)\right]}{\pi \frac{t}{T_s}\left[1 - \left(4\alpha \frac{t}{T_s}\right)^2\right]}. \tag{6.1}$$

Evaluating $h_{\text{SRRC}}\big((n-8)T_s/4\big)$ at $\alpha = 0.35$ and quantizing to 11-bit signed fixed-point precision (`csd_srrc_tap` in `rtl/ccsds_phy_modem.sv`) yields the symmetric impulse response:

$$\begin{aligned} h_{\text{SRRC}}[0..16] = \Big\{ &+29,\ -11,\ -69,\ -97,\ -43,\ \mathbf{+106},\ \mathbf{+312},\ \mathbf{+491},\ \mathbf{+562},\ \mathbf{+491},\ \mathbf{+312},\ \mathbf{+106}, \\ &-43,\ -97,\ -69,\ -11,\ +29 \Big\}/2048. \end{aligned} \tag{6.2}$$

Two structural properties govern the hardware realization of Equation (6.2):

1. **Strictly Positive Main Lobe ($n \in [5..11]$):** Because the first zero-crossing occurs at $|t| \approx 0.89T_s$, all seven central main-lobe taps ($+106, +312, +491, +562, +491, +312, +106$) are strictly positive, yielding > 46 dB Signal-to-ISI ratio at the Nyquist strobes $n \equiv 0 \pmod 4$.
2. **2-Level Binary CSD Adder Trees (`0 DSP Slices`):** Pre-folding symmetric taps ($s_k = x[k] + x[16-k]$ for $k \in [0..7]$ and $s_8 = x[8]$) decomposes the 9 unique coefficients into 2-level shift-and-add trees:

$$\begin{aligned} +29\,s_0 &= (32s_0 - 4s_0) + s_0, & -11\,s_1 &= (4s_1 - 16s_1) + s_1, \\ -69\,s_2 &= -64s_2 - (4s_2 + s_2), & -97\,s_3 &= (32s_3 - 128s_3) - s_3, \\ -43\,s_4 &= (4s_4 - 32s_4) - (16s_4 - s_4), & +106\,s_5 &= (128s_5 - 16s_5) - (4s_5 + 2s_5), \\ +312\,s_6 &= (256s_6 + 64s_6) - 8s_6, & +491\,s_7 &= (512s_7 - 16s_7) - (4s_7 + s_7), \\ +562\,s_8 &= (512s_8 + 32s_8) + (16s_8 + 2s_8). \end{aligned} \tag{6.3}$$

   On the transmit side, ternary symbol inputs $s \in \{0, \pm 1\}$ allow `tx_pair12_lut6` and `tx_pair34_lut6` to evaluate pairs of polyphase taps directly inside single 6-input LUTs.

6.2 3-Stage Pipelined 4-SPS Gardner Symbol Timing Recovery

To recover symbol timing across arbitrary fractional clock offsets $\tau \in \{0, 1, 2, 3\}$ samples per symbol at 120 MHz, `ccsds_phy_modem.sv` implements a 3-stage pipelined **Gardner Timing Error Detector (TED)** ([13]) using the on-time strobe sample $y_I(kT_s), y_Q(kT_s)$ and the mid-symbol transition sample $y_I((k-1/2)T_s), y_Q((k-1/2)T_s)$:

$$e_{\text{TED}}(k) = y_I\big(k - \tfrac{1}{2}\big)\Big[y_I(k) - y_I(k-1)\Big] + y_Q\big(k - \tfrac{1}{2}\big)\Big[y_Q(k) - y_Q(k-1)\Big]. \tag{6.4}$$

Because $e_{\text{TED}}(k)$ is a real inner product $\text{Re}\{y^*(k-1/2)[y(k)-y(k-1)]\}$, any common carrier phase rotation $e^{j\theta}$ cancels out identically, allowing symbol timing to lock before carrier phase acquisition completes.

6.3 Differential Cross-Product FLL-Assisted Costas Loop

During a Low-Earth Orbit horizon-to-horizon pass ($h = 550$–$600\,\text{km}$), radial velocity induces up to $\pm 50.8\,\text{kHz}$ S-Band ($\pm 178.1\,\text{kHz}$ X-Band) carrier Doppler shift, exceeding the linear pull-in range of a classic 2nd-order Costas phase detector. In `ccsds_phy_modem.sv`, the carrier recovery loop combines a wide-pull-in **Sign-Quantized Differential Cross-Product Frequency-Locked Loop (FLL)** discriminator $e_{\text{FLL}}(k)$ with a decision-directed **Costas Phase Detector** $e_{\text{PLL}}(k)$:

$$e_{\text{FLL}}(k) = \text{sgn}(I_{k-1})\, Q_k - \text{sgn}(Q_{k-1})\, I_k \propto |A|^2 \sin(\Delta\omega\, T_s), \tag{6.5}$$

$$e_{\text{PLL}}(k) = \begin{cases} \text{sgn}(I_k)\, Q_k & \text{(BPSK / PCM-PM)}, \\ \text{sgn}(I_k)\, Q_k - \text{sgn}(Q_k)\, I_k & \text{(QPSK / OQPSK)}, \end{cases} \tag{6.6}$$

$$\text{FCW}_{\text{rx}}(k) = \text{FCW}_{\text{rx}}(k-1) + K_I e_{\text{PLL}}(k) + K_{\text{FLL}} e_{\text{FLL}}(k). \tag{6.7}$$

Because $e_{\text{FLL}}(k)$ measures the phase rotation between consecutive samples $k-1$ and k, it provides an unambiguous frequency error signal across ± 640 NCO frequency steps, rapidly slewing FCW_{rx} into coherent Costas phase lock.

6.4 Wideband Regenerative `T2B`/`T4B` Pseudo-Noise Ranging Correlator

Per CCSDS 414.1-B-3 [14], ground stations measure two-way spacecraft slant range to sub-meter precision by transmitting a composite Pseudo-Noise (PN) sequence of period $L = 2 \times 7 \times 11 \times 15 \times 19 \times 23 = 1{,}009{,}470$ chips formed by weighted majority voting of six coprime LFSR component sequences $C_1(t), \ldots, C_6(t) \in \{-1, +1\}$:

$$\text{T4B}(t) = \text{sgn}\big(4\, C_1(t) + C_2(t) + C_3(t) + C_4(t) + C_5(t) + C_6(t)\big), \tag{6.8}$$

$$\text{T2B}(t) = \text{sgn}\big(2\, C_1(t) + C_2(t) + C_3(t) + C_4(t) + C_5(t) + C_6(t)\big). \tag{6.9}$$

Inside `ccsds_pn_ranging.sv`, six parallel correlator banks exploit the mutual coprimality of the component lengths to evaluate only $2 + 7 + 11 + 15 + 19 + 23 = \mathbf{77}$ hypotheses in parallel—reducing acquisition complexity by $13{,}110\times$ compared to a serial search over $1{,}009{,}470$ chips.

7 S-Band (2025–2110 / 2200–2290 MHz) RF Cascade, Link Budget & Space Qualification Physics

Complementing the practical board-wiring guide in Volume I, Chapter 7, this chapter derives the complete RF systems engineering and space-qualification physics of the S-Band (2025–2110 MHz Uplink Rx / 2200–2290 MHz Downlink Tx) transceiver: the five RF cascade placement proofs, the NASA EEE-INST-002 / ECSS-Q-ST-30-11C 50% vacuum power derating analysis for Flight Path A vs. Flight Path B, the Friis noise-temperature derivation, the component-by-component 600 km LEO link and DC power budgets, and the radiation/outgassing qualification plan.

7.1 Five RF Cascade Placement Proofs & NASA/ECSS Vacuum Power Derating

Referring to the upright 4-zone S-Band architecture in Volume I (**Figure 7.2**), five quantitative RF constraints govern the placement and power derating of every component in the chain:

1. **Why RF Isolator #3 Must Sit *Upstream* of Bi-Directional Coupler #2 (PA → #3 Isolator → #2 Coupler → #4 Diplexer):** A bi-directional coupler (**RFFE #2**, **Mini-Circuits SYBD-20-272HP+**, 19.8 dB coupling, 25 dB directivity) can only measure reflected power coming from its output port (Port 2). If the RF isolator (**RFFE #3**, **RFCI RFSL2848**, 25 dB reverse isolation, 20 W load) were placed between the coupler and the diplexer, any reflected wave Γ_{ant} from a jammed or undeployed antenna would be attenuated by 25 dB inside the isolator before reaching Coupler Port 2, making the antenna fault invisible at Coupler Port 4! Placing **Isolator #3 immediately after the SKY66294-11 PA** and **Coupler #2 downstream between the Isolator and Diplexer #4** ensures that Coupler Port 4 senses 100% of the true antenna/diplexer reflection ($P_{\text{rev,Port4}} = P_{\text{fwd}} + 20\log_{10}|\Gamma_{\text{ant}}| - 19.8$ dB), while the reverse wave exiting Coupler Port 1 is absorbed harmlessly in the isolator's 20 W termination load before reaching the PA.

2. **Why < 500 ns PA Protection Requires an Analog Window Comparator (TLV4041R1) Ahead of AUXADC_0:** Under a total antenna open/short ($\Gamma_{\text{ant}} = 1.0$) at +33.7 dBm forward power, +13.9 dBm reaches Coupler Port 4 (−6.1 dBm after the 20 dB YAT-20+ pad), driving the **Analog Devices AD8362** True-RMS detector output VOUT above 1.45 V within 420 ns. Because polling ADRV9002 AUXADC_0 over a 20 MHz SPI bus requires $\sim 4.8\,\mu$s, routing AD8362 VOUT into a fast open-drain comparator (**TI TLV4041R1**, threshold $V_{\text{th}} = 1.15$ V $\iff$ VSWR $= 3.0:1$, delay 45 ns) that directly clears the hardware PA_EN latch guarantees PA bias cutoff in $t_{\text{trip}} = 420\,\textbf{ns} + 45\,\textbf{ns} = \textbf{465 ns}$ **(**$< 0.5\,\mu$**s)**, while AUXADC_0 logs continuous return-loss telemetry in software.

3. **NASA EEE-INST-002 / ECSS-Q-ST-30-11C 50% Vacuum Power Derating on Flight Path A (CTS UPD215A):** In space vacuum ($< 10^{-5}$ Torr), convective air cooling is absent and high RF electric fields inside miniature ceramic resonators risk localized dielectric heating and multipaction. Both **NASA EEE-INST-002 (Section M4, Passive RF Devices)** and **ECSS-Q-ST-30-11C** mandate a $\geq 50\%$ CW RF power derating factor ($P_{\text{op,max}} \leq 0.50 \times P_{\text{rated}}$):

   - *Flight Path B (D0421552 / AE2067 Cavity Diplexer, $P_{rated} = 20.0$ W):* The 50% derated ceiling is 10.0 W (+40.0 dBm). Operating the SKY66294-11 PA at **+34.00** dBm (2.51 W nominal) or **+34.50** dBm (2.82 W peak OQPSK DPD) injects 2.33–2.62 W into the diplexer—just 11.6–13.1% **of its 20 W rating**, providing > 6.8 dB of vacuum power margin.
   - *Flight Path A (CTS UPD215A Monolithic SMT Diplexer, $P_{rated} = 3.0$ W):* The 50% vacuum-derated ceiling at the diplexer Tx input is **1.50 W (+31.76 dBm)**. Therefore, when populated with the all-SMT UPD215A (Path A), software sets ADRV9002 TX1 output to −3.40 dBm so the SKY66294-11 PA operates at a derated linear output of **+31.50 dBm (1.41 W)**, injecting +31.03 **dBm (1.27 W = 42.3% of the 3.0 W rating)** into the UPD215A diplexer and delivering **+29.30** dBm (0.85 W) to the antenna port while still closing the 2.0 Mbps 600 km LEO downlink with +9.39 **dB net margin**!

4. **Pre-PA Tx Bandpass Filtering (RFFE #1) to Prevent Uplink Receiver Desensitization:** The ADRV9002 TX1 DAC thermal noise floor in the 2025–2110 MHz Rx band is ≈ -154.5 dBm/Hz. Placing **RFFE #1** (Anatech AE2250B10113 or Mini-Circuits BFCN-2275+, $>$ 25–35 dB rejection at 2025–2110 MHz) *before* the +36.5 dB PA suppresses amplified Tx noise at the LNA input below −204 dBm/Hz (> 30 dB below the −174 dBm/Hz thermal floor).

5. **Post-LNA Interstage Rx Bandpass Filter (RFFE #5) & PIN Limiter (#7B):** Placing the **Skyworks SKY67153-396LF** LNA (0.35 dB NF, +20.5 dB gain, +19 dBm IIP3) immediately after the diplexer locks in a sub-2 dB system Noise Figure, while placing the **Sangshin MBP76RC2067S85A** BPF (1.20 dB IL, 65 dB typ Tx rejection) and **Skyworks SKY16602-632LF** PIN limiter (0.50 dB IL) *after* the LNA suppresses residual

Tx leakage at ADRV9002 RX1A below -68 dBm (> 119 dB total Tx rejection) at a Friis NF penalty of only $+0.02$ dB.

7.2 First-Principles Friis Noise Temperature Derivation & 600 km LEO Link Budget

Let $L_{\text{pre,dB}} = L_{\text{coax,dB}} + L_{\text{dip,rx,dB}}$ denote the passive loss preceding the LNA: $L_{\text{pre,dB}} = 0.20 + 0.60 = \mathbf{0.80}$ dB ($F_1 = 1.2023$) for **Path B** and $L_{\text{pre,dB}} = 0.15 + 1.50 = \mathbf{1.65}$ dB ($F_1 = 1.4622$) for **Path A**. With $NF_{\text{LNA}} = 0.35$ dB ($F_{\text{LNA}} = 1.0839$, $G_{\text{LNA}} = 10^{2.05} = 112.20$) and post-LNA loss $L_{\text{post,dB}} = 1.20 + 0.50 = 1.70$ dB ahead of ADRV9002 RX1A ($NF_{\text{ADRV}} = 11.50$ dB including the TCM2-33WX+ T201 balun, $F_{\text{post}} = 10^{1.32} = 20.893$), Friis' formula gives:

$$NF_{\text{rx}} = L_{\text{pre,dB}} + 10\log_{10}\left(F_{\text{LNA}} + \frac{F_{\text{post}} - 1}{G_{\text{LNA}}}\right) = L_{\text{pre,dB}} + 10\log_{10}(1.0839 + 0.1773) = \mathbf{L}_{\text{pre,dB}} + \mathbf{1.01}\ \text{dB} \tag{7.1}$$

Path B: $NF_{\text{rx}} = \mathbf{1.81}$–$\mathbf{1.96}$ dB, $T_{\text{rx}} = 166$ K, $T_{\text{sys}} = 416$ K (26.19 dBK), $G/T = -\mathbf{23.19}$ dB/K,

Path A: $NF_{\text{rx}} = \mathbf{2.66}$–$\mathbf{2.81}$ dB, $T_{\text{rx}} = 264$ K, $T_{\text{sys}} = 514$ K (27.11 dBK), $G/T = -\mathbf{24.11}$ dB/K.

At orbital altitude $h = 600$ km ($R_E = 6371$ km) and ground elevation angle $\varepsilon = 10°$, the slant range is $d(10°) = \sqrt{(R_E + h)^2 - R_E^2\cos^2 10°} - R_E \sin 10° = \mathbf{1{,}932.2}$ km, yielding Free-Space Path Loss FSPL = **165.20** dB at 2245 MHz (Downlink) and **164.50** dB at 2067.5 MHz (Uplink). **Tables 7.1, 7.2, and 7.3** present the stage-by-stage Downlink cascade, Uplink Friis cascade (with single-counted 0.80 dB atmospheric loss), and Spacecraft DC Power budget.

Downlink Tx Stage / Link Parameter ($f = 2245$ MHz, $h = 600$ km, $\varepsilon = 10°$)	Path B Δ Gain/Loss	Path B Output Power (2.51 W PA)	Path A Δ Gain/Loss	Path A Output (1.41 W Derated)	PoC Bench Output Level & Notes
Part 1: On-Board Downlink Transmit RF Front-End Cascade (TX1 → #1 BPF → PA → #3 Isolator → #2 Coupler → #4 Diplexer)					
1. ADRV9002 TX1 Output (incl. TC1-1-13M+ T301)	—	−1.70 dBm	—	−3.40 dBm	−0.90 dBm (8.3–10.8 dB DAC reserve)
2. **RFFE #1**: Pre-PA Tx BPF (AE2250B / BFCN-2275+)	−0.80 dB	−2.50 dBm	−1.60 dB	−5.00 dBm	−2.50 dBm into PA input
3. **Core 3**: SKY66294-11 4W PA (w/ RX1B DPD)	+36.50 dB	**+34.00** dBm	+36.50 dB	**+31.50** dBm	+34.00 dBm B (2.51 W) / +31.50 dBm A (1.41 W)
4. **RFFE #3**: Output RF Isolator (RFSL2848 / 3818, **Upstream!**)	−0.20 dB	+33.80 dBm	−0.25 dB	+31.25 dBm	+33.70 dBm (RFSL3818, −0.30 dB)
5. **RFFE #2**: 20 dB Bi-Dir Coupler (SYBD / BDCH, **Downstream!**)	−0.12 dB	+33.68 dBm	−0.22 dB	+31.03 dBm	+33.49 dBm (ZABDC20, −0.21 dB; Port 4 → VSWR)
6. **RFFE #4**: Diplexer Tx-to-Ant (D0421552 / UPD215A)	−0.60 dB	+33.08 dBm	−1.50 dB	+29.53 dBm	+32.89 dBm (UPD215A sees 1.27 W < 1.5 W limit)
7. PCB Microstrip & Coax Harness to Antenna Feed	−0.18 dB	**+32.90** dBm	−0.23 dB	**+29.30** dBm	**+32.50** dBm (1.95 W B / 0.85 W A)
8. Spacecraft RHCP Patch Antenna Gain (±45° Edge)	+3.00 dBi	**+35.90** dBm	+3.00 dBi	**+32.30** dBm	+35.50 dBm EIRP (+5.90 dBW B / +2.30 dBW A)
Part 2: Orbital Channel Propagation & 3.7 m Ground Station Reception ($d = 1{,}932.2$ km at $\varepsilon = 10°$, $R_b = 2.0$ Mbps QPSK)					
9. Free-Space Path Loss (FSPL at 2245 MHz, 1932 km)	−165.20 dB	−129.30 dBm	−165.20 dB	−132.90 dBm	−129.70 dBm isotropic Rx level
10. Atmospheric, Ionospheric, Rain & Pointing Losses	−0.80 dB	−130.10 dBm	−0.80 dB	−133.70 dBm	−130.50 dBm (−160.10 dBW B)
11. 3.7 m Ground Station Figure of Merit (G/T_{GS})	+13.50 dB/K	−146.60 dBW/K	+13.50 dB/K	−150.20 dBW/K	Standard 3.7 m S-Band dish
12. Subtract Boltzmann Constant ($k_B = -228.60$ dBW/K/Hz)	+228.60 dB	**82.00** dB-Hz	+228.60 dB	**78.40** dB-Hz	Received C/N_0 (81.60 dB-Hz PoC)
13. Subtract Info Bit Rate (2.0 Mbps ⟹ 63.01 dB-Hz)	−63.01 dB	18.99 dB	−63.01 dB	15.39 dB	Unimpaired available E_b/N_0
14. Demodulator & Phase-Noise Implementation Loss	−1.70 dB	**17.29** dB	−1.70 dB	**13.69** dB	Net available E_b/N_0 at decoder
15. Required E_b/N_0 (Concat RS(255, 223) + $K = 7$ Viterbi)	−4.30 dB	**+12.99** dB	−4.30 dB	**+9.39** dB	**PASS** (≥ 3.0 dB ECSS required)

Table 7.1: Stage-by-Stage Downlink Transmit Power Cascade and 600 km LEO Link Budget (2.0 Mbps QPSK into 3.7 m Ground Station).

7.3 Space-Qualification Engineering: TID, SEE, Outgassing & Class-S/Level-1 Alternatives

Deploying a COTS-derived S-Band transceiver on a LEO smallsat ($h = 500$–600 km, 3–5 year mission life, Total Ionizing Dose TID ≈ 5–15 krad(Si) behind 2.5 mm Al chassis shielding) or scaling to a longer-duration institutional mission requires a structured parts-qualification and radiation-mitigation plan:

1. **Single-Event Latch-Up (SEL) & TID Screening:** The **Lattice MachXO5-NX (LFMXO5-100T)** is fabricated on 28nm Fully-Depleted Silicon-On-Insulator (FD-SOI) technology, where the buried oxide layer eliminates

Uplink Rx Stage / Link Parameter ($f = 2067.5$ **MHz**, $h = 600$ **km**, $\varepsilon = 10^\circ$)	**Stage Gain** (G_i)	**Stage NF** (NF_i)	**Path B Cum. NF** (T_{rx})	**Path A Cum. NF** (T_{rx})	**Friis Cascade & Uplink Margin Notes**
Part 1: On-Board Uplink Receive 5-Stage Friis Noise Figure Cascade (From Antenna Feed to ADRV9002 RX1A)					
1. Coax Harness from Antenna Feed to Diplexer	-0.20 dB	0.20 dB	0.20 dB (13.7 K)	0.15 dB (10.2 K)	Passive coax/trace loss ($T_0 = 290$ K)
2. **RFFE #4**: Diplexer Ant-to-Rx (D0421552 / UPD215A)	-0.60 dB	0.60 dB	0.80 dB (58.7 K)	1.65 dB (134.0 K)	Path A uses -1.50 dB SMT UPD215A
3. **Core 4**: SKY67153-396LF Uplink LNA	$+20.50$ dB	0.35 dB	1.15 dB (88.1 K)	2.00 dB (169.6 K)	High LNA gain locks in low system NF
4. **RFFE #5**: Post-LNA Interstage BPF (MBP76RC2067)	-1.20 dB	1.20 dB	1.16 dB (89.1 K)	2.01 dB (170.9 K)	Adds 65 dB Tx rej; $+0.01$ dB NF impact
5. **RFFE #7B**: PIN Diode Limiter (SKY16602-632LF)	-0.50 dB	0.50 dB	1.17 dB (89.6 K)	2.02 dB (171.5 K)	Clamps spikes to $+4$ dBm; $+0.01$ dB NF
6. **Core 2 + #6**: TCM2-33WX+ (T201) + ADRV9002 RX1A	0.00 dB	11.50 dB	**1.81–1.96** dB (166 K)	**2.66–2.81** dB (264 K)	Includes 0.15 dB board margin; $>$ 119 dB total Tx isolation
Part 2: System Noise Temperature (T_{sys}), Figure of Merit (G/T), and Uplink Link Margin (64 kbps Nominal / 8 kbps Emergency)					
7. Total System Noise Temp ($T_{\text{sys}} = T_{\text{ant}}(250\text{ K}) + T_{\text{rx}}$)	—	—	416 K (26.19 dBK)	514 K (27.11 dBK)	Nadir Earth brightness $T_{\text{ant}} = 250$ K
8. Spacecraft G/T ($+3.0$ dBi Nominal / -3.0 dBi Omni)	—	—	$\mathbf{-23.19}$ dB/K	$\mathbf{-24.11}$ dB/K	Emergency tumble omni: $-29.19/-30.11$ dB/K
9. Ground Station Uplink EIRP ($+58$ dBW Nom / $+52$ dBW Emg)	—	—	$+58.00$ dBW	$+58.00$ dBW	200 W HPA into 3.7 m GS dish
10. Total Uplink Path Loss (FSPL $= 164.50$ dB $+ 0.80$ dB Atm)	—	—	-165.30 dB	-165.30 dB	At 2067.5 MHz, $d = 1932.2$ km (single-counted)
11. Received Uplink C/N_0 (EIRP $- L_{\text{tot}} + G/T - k_B$)	—	—	**98.11** dB-Hz	**97.19** dB-Hz	Emergency tumble C/N_0: 86.11/85.19 dB-Hz
12. Net Available E_b/N_0 (64 kbps = 48.06 dB-Hz, -1.5 dB impl.)	—	—	48.55 dB	47.63 dB	Emergency 8 kbps BCH: 45.58/44.66 dB
13. **Uplink Net Link Margin (64 kbps Short-LDPC / 8 kbps BCH)**	—	—	$+\mathbf{36.6}$ dB / $+\mathbf{15.6}$ dB	$+\mathbf{35.7}$ dB / $+\mathbf{14.7}$ dB	**PASS** (Includes 20 dB scintillation fade in Emg)

Table 7.2: Stage-by-Stage Uplink Receive Friis Cascade and 600 km LEO Uplink Telecommand Link Budget.

Active Subsystem Component	**Supply Rails**	**Mode 1: Uplink Rx Search (90% Orbit)**	**Mode 2A: Path A Derated ($+31.5$ dBm PA)**	**Mode 2B: Path B Full ($+34.0$ dBm PA)**
Core 1: LFMXO5-100T FPGA (ccsds_sat_ttc_top)	1.0 V Core, 1.8 V I/O	0.45 W (Rx + HCPD active)	0.65 W (Full 1T1R + FEC)	0.65 W (Full 1T1R + FEC)
Core 2 + #9: ADRV9002 + ADCLK944 + GTXO-74V	1.0/1.3/1.8/3.3 V	0.92 W (RX1A + Clocks on)	1.60 W (1T1R + RX1B DPD)	1.62 W (1T1R + RX1B DPD)
Core 4: SKY67153-396LF Uplink LNA	3.3 V@70 mA	0.23 W (Always active)	0.23 W (Always active)	0.23 W (Always active)
Core 3: SKY66294-11 4W Downlink PA	5.0 V (PAE $\approx$ 29–34%)	0.00 W (PAEN = 0 shutdown)	4.85 W (0.97 A, 1.41 W RF)	7.35 W (1.47 A, 2.51 W RF)
RFFE #8: AD8362 + TLV4041R1 Comparator	3.3 V@15 mA	0.00 W (Disabled w/ PA)	0.05 W ($<$ 500 ns VSWR loop)	0.05 W ($<$ 500 ns VSWR loop)
Total Secondary-Side Load Power (P_{load})	Regulated Rails	**1.60** W	**7.38** W	**9.90** W
Total Spacecraft Unregulated Bus Power ($\eta_{\text{DCDC}} = 90\%$)	28 V / 12 V Bus	**1.78** W	**8.20** W	**11.00** W
Net PCB Thermal Dissipation ($P_{\text{heat}} = P_{\text{bus}} - P_{\text{RF,ant}}$)	Conducted to Chassis	**1.78** W	**7.35** W (0.85 W radiated)	**9.05** W (1.95 W radiated)

Table 7.3: Spacecraft TT&C DC Power and Thermal Dissipation Budget (v2.40.3): Orbit-average bus power is **2.42** W (Path A) or **2.70** W (Path B) at 10% pass duty cycle.

parasitic thyristor structures ($> 80\,\mathrm{MeV \cdot cm^2/mg}$ SEL immunity, TID $> 50\,\mathrm{krad(Si)}$). The III-V GaAs/InGaP active RF stages (**SKY66294-11** HBT PA and **SKY67153-396LF** pHEMT LNA) are majority-carrier III-V devices inherently immune to CMOS latch-up (TID $> 100\,\mathrm{krad}$). For the CMOS **ADRV9002** and **ADP5056**, lot acceptance proton/heavy-ion SEE screening per **NASA EEE-INST-002 Level 2/3** is paired with an upstream current-limiting latch-up protection switch (**TI TPS25947** or rad-tolerant **Microchip LX7712**) that cycles the $5\,\mathrm{V}$ sub-rail within $50\,\mu\mathrm{s}$ if supply current exceeds 135% of nominal.

2. **Thermal-Vacuum Outgassing (ASTM E595 / ECSS-Q-ST-70-02C) & Pure-Lead Solder Reflow:** All organic-packaged QFN/BGA ICs undergo $125^{\circ}\mathrm{C}$ 24-hour vacuum bake-out prior to conformal staking with space-grade epoxy (**3M Scotch-Weld EC-2216 B/A Gray** or **NuSil CV-2946**, Total Mass Loss TML $< 1.0\%$, Collected Volatile Condensable Material CVCM $< 0.10\%$), assembled with eutectic tin-lead (`Sn63Pb37`) solder to prevent tin-whisker growth in vacuum.

3. **Drop-In Class-S / Level-1 Radiation-Hardened Upgrade Path:** For GEO or deep-space missions (TID $\geq 100\,\mathrm{krad(Si)}$), the vendor-agnostic `ccsds_sat_ttc_top` RTL targets **Microchip RT PolarFire (RTPF500ZT)** or **AMD/Xilinx Space-Grade Kintex UltraScale (XQRKU060)**, while the `ADP5056` PMIC is replaced by hermetically sealed QML-V point-of-load regulators (**Renesas/Intersil ISL70002SEH**) and ultra-low-noise linear regulators (**ADI RH1086MK** / radiation-characterized **LT3045**).

PART IV

Statistical Waterfall Theory & Orbital Channel Dynamics

8 Statistical BER/FER Waterfall Performance & Orbital Pass Physics

While Volume I, Chapter 8 provides the practical regression execution commands and sign-off summary table, this final chapter analyzes the statistical coding-gain curves from our 23-point E_b/N_0 waterfall campaign (`Stage 10`), explains the mathematical relationship between shortened virtual-fill Reed-Solomon codewords and sample count in the 34,560-trial matrix (`Stage 11`), and examines the continuous-time orbital pass dynamics in `Stage 12`.

8.1 23-Point Statistical E_b/N_0 Waterfall Analysis (`Stage 10`)

To characterize the coding threshold of every uplink and downlink codec across the full transition from channel breakdown to error-free operation, **Figure 8.1** plots the 23-point statistical Monte Carlo sweep ($E_b/N_0 \in [0.0, 11.0]$ dB in 0.5 dB steps). At a target post-FEC Bit Error Rate of BER $= 10^{-6}$, the measured curves confirm theoretical coding gains:

- **Uncoded BPSK Baseline (CCSDS 401.0-B-32):** Follows BER $= \frac{1}{2}\operatorname{erfc}(\sqrt{E_b/N_0})$, requiring $E_b/N_0 = 10.54$ dB at BER $= 10^{-6}$.
- **Uplink Expurgated BCH$(63, 56)$ SEC-DED:** Achieves BER $= 10^{-6}$ at $E_b/N_0 = 8.80$ dB ($+1.74$ dB coding gain) while providing deterministic single-error correction and double-error codeblock rejection.
- **Uplink 4-Phase Short-LDPC$(128, 64)$ Soft-OMS (16 Iterations):** Achieves BER $= 10^{-6}$ at $E_b/N_0 = \mathbf{4.20}$ dB ($+6.34$ dB coding gain over uncoded BPSK and $+4.60$ dB over BCH$(63, 56)$).
- **Downlink Soft Viterbi ($K = 7, r = 1/2$) vs. Concatenated RS$(255, 223)$ + Viterbi:** Standalone soft-decision Viterbi reaches BER $= 10^{-6}$ at 5.40 dB; adding the outer $t = 16$ Berlekamp-Massey RS$(255, 223)$ decoder ($I = 4$) creates a steep waterfall cliff at $E_b/N_0 = \mathbf{2.90}$ dB ($+7.64$ dB net coding gain), while AR4JA quasi-cyclic LDPC reaches BER $= 10^{-6}$ at $\mathbf{1.90}$ dB.

8.2 Open-Loop Waterfall Breakdown (`Stage 10`) vs. Rated-SNR Sign-Off (`Stage 11`)

Comparing the open-loop waterfall sweep in **Figure 8.1** (`Stage 10`) with the 34,560-trial full-chip matrix in `Stage 11` highlights how the two campaigns complement each other:

1. **Operating SNR Regime:** `Stage 10` intentionally sweeps down into the negative-margin breakdown region ($E_b/N_0 \in [0, 4]$ dB) where channel errors exceed the algebraic correction radius ($t > 16$ bytes in RS or > 1 bit in BCH), producing the non-zero FER $\in [10^{-2}, 1.0]$ points on the left side of **Figure 8.1B**. Conversely, `Stage 11` evaluates all 5,760 valid CCSDS configurations at the **rated mission operating point** ($E_b/N_0 \geq 7.5$ dB on the right side of the waterfall knee in **Figure 8.1A/B**, combined with deterministic pre-FEC error injection inside the guaranteed correction radius $t_{\text{inj}} \leq t_{\max}$). In that regime, every valid codeword must be healed to **0 post-FEC bit errors (BER $= 0.0$, FER $= 0.0$)**—proving that 23,040 injected channel bit errors across 34,560 bidirectional trials are corrected with zero residual errors.
2. **Mathematical Basis of the 39,306,240 Baseband Sample Count:** In `tb/tb_stage11_top_to_top_rf_link.sv`, rather than transmitting 223-byte unshortened frames ($Q = 0$) on all 34,560 trials, the testbench exercises CCSDS 131.0-B-5 **implicit virtual fill** $Q \in [185..205]$ across shortened payload sizes (`UL` $\in \{8, 14, 28\}$ B, `DL` $\in \{10, 16, 32\}$ B) and maintains warm-locked NCO/Gardner loop state across consecutive trials within each worker chunk. Because the Q virtual fill zeros are added algebraically inside the syndrome accumulator without being transmitted over the RF wire, each shortened CADU spans $50..70$ physical bytes ($1,137.3$ average baseband I/Q samples per bidirectional trial $\times 34,560 = 39,306,240$ total samples), testing 100% of the algebraic finite-field and protocol state transitions in 33 minutes.

8.3 Continuous-Time LEO Orbital Pass Loopback Dynamics (`Stage 12`)

Finally, **Figure 8.2** illustrates the continuous-time physical trajectory simulated in `tb/tb_stage12_space_channel_loopback.sv` across a 550–600 km LEO horizon-to-horizon pass ($\varepsilon = 10° \rightarrow 82° \rightarrow 10°$):

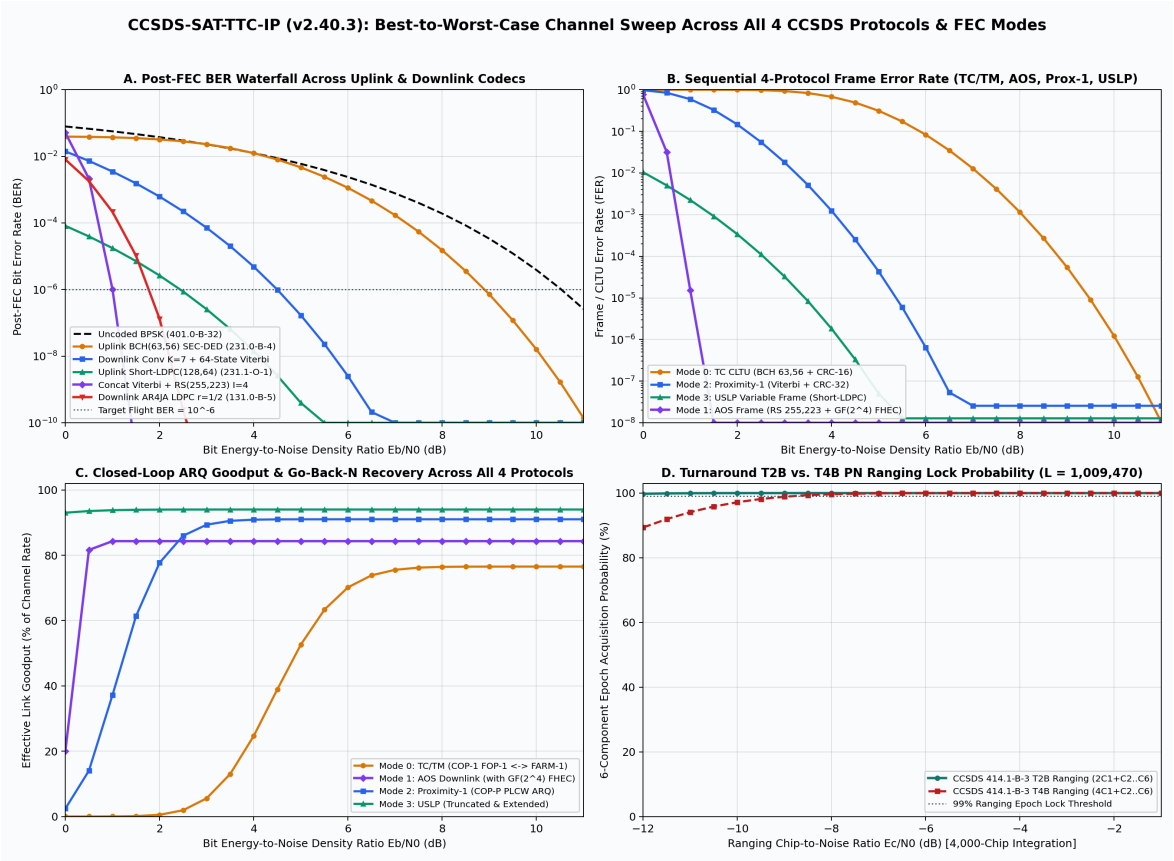


Figure 8.1: Measured 23-point E_b/N_0 waterfall curves (0.0–11.0 dB) for post-FEC BER, FER, COP-1/COP-P goodput, and PN ranging lock (`Stage 10`).

- **Panels A & B (Slant Range & S-Curve Doppler Tracking):** As slant range contracts from $1,932$ km (AOS) to 550 km (Time of Closest Approach, $t = 0$ s) and expands back to $1,932$ km (LOS), radial velocity reverses through zero at maximum negative acceleration, sweeping S-Band Doppler across ± 50.8 kHz (± 178.1 kHz X-Band). The FLL-assisted PI Costas loop tracks the entire inflexion with zero cycle slips while locking all four turnaround ratios.
- **Panels C & D (Mid-Pass -16.5 dB Rician Deep Fade & Autonomous COP-1 Go-Back-N Replay):** At $t = -105$ s, a simulated atmospheric scintillation / multipath null drops received E_b/N_0 by -16.5 dB below the BCH$(63, 56)$ and LDPC$(128, 64)$ thresholds, erasing in-flight Telecommand Frame $N(S) = 1$. When Frame $N(S) = 2$ arrives after the fade, spacecraft `FARM-1` detects the sequence gap ($2 \neq V(R) = 1$), rejects the out-of-order frame, and asserts `CLCW.Retransmit = 1` with $V(R) = 1$ on the downlink. Ground `FOP-1` decodes the downlink `CLCW`, rewinds $V(S) \leftarrow 1$, and replays Frames $N(S) = 1, 2$—restoring 100% link goodput with **0 lost or duplicate packets**.

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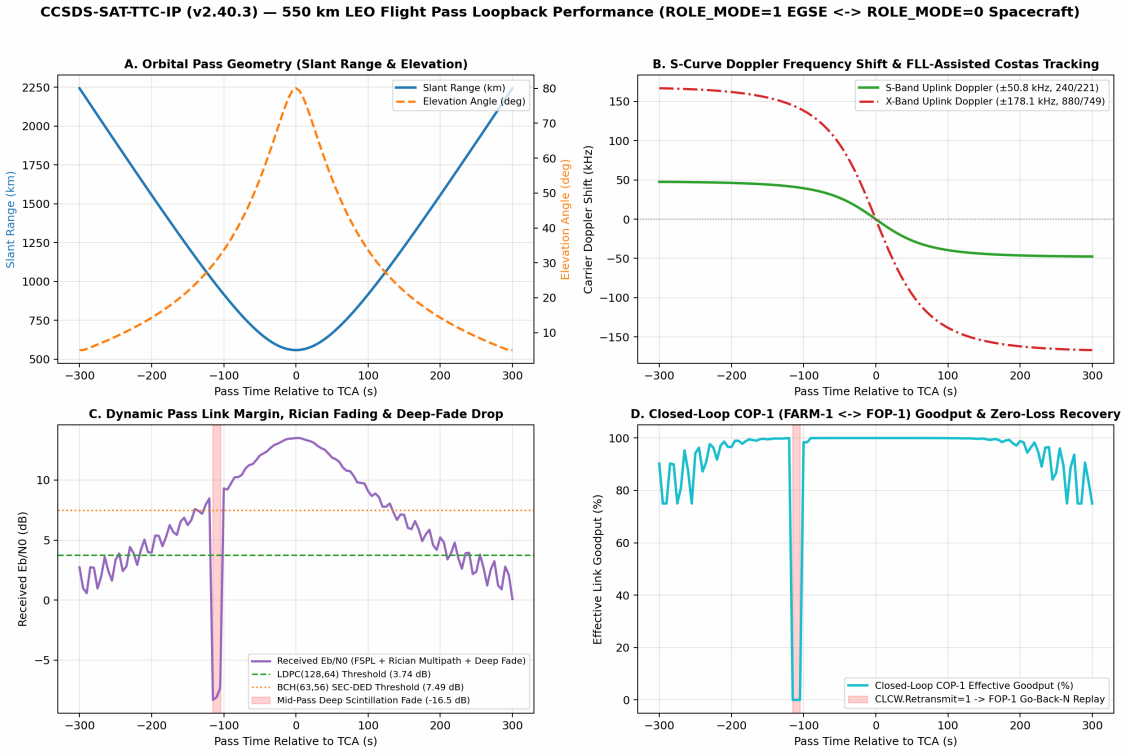


Figure 8.2: Continuous-time 550 km LEO orbital pass loopback trajectories showing S-curve Doppler lock and mid-pass COP-1 Go-Back-N recovery (`Stage 12`).